



Determinants of job satisfaction in the UAE

A case study of the Dubai police

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Abstract

Purpose – The aim of this study is to identify the factors affecting job satisfaction in the UAE. Specifically, the study seeks to examine the relative effects of demographic and environmental factors on job satisfaction among Dubai police employees.

Design/methodology/approach – A scale development process was used. The first stage was a review of the literature on the determinants of job satisfaction in order to generate a pool of items that reflect the job satisfaction construct. A list of items was gathered to cover all aspects of job satisfaction. In-depth interviews and a focus group were also used to generate additional items.

Findings – Overall, the findings show that, in a collectivist culture such as the UAE, both intrinsic and extrinsic factors can be a source of job satisfaction or dissatisfaction.

Originality/value – Given the importance of understanding the determinants of job satisfaction, and the paucity of research on the topic in the Middle East, this study is of great importance to both scholars and practitioners in the region.

Keywords Job satisfaction, Determinants, Police, United Arab Emirates

Paper type Literature review

Introduction

This paper examines the role that demographic and environmental factors play in determining job satisfaction in the United Arab Emirates (UAE). Demographic factors are attributes of the individual such as age, race, gender, education level, and years of work experience. Environmental factors are characteristics of the immediate job environment such as the skills variety required to carry out the job, task significance, autonomy, and interaction with co-workers. Although more studies have been undertaken to explore the antecedents and consequences of job satisfaction than any other variables in organisations (Spector, 1997), the topic has largely escaped research attention in the Middle East. Results from past studies in Western countries cannot be applied directly to Middle Eastern organizations. An extensive body of literature has reported variance in the determinants and consequences of job satisfaction across cultures (Spector, 2008; Kirkman and Shapiro, 2001; Diener *et al.*, 2003; Vecernik, 2003; Kanungo and Wright, 1983; Lincoln and Kalleberg, 1985; Huang and Van de Vliert, 2004; Diener *et al.*, 1995). Taken together, these studies urge scholars to consider



cultural contexts when examining the determinants and consequences of job satisfaction. Therefore, given the importance of understanding the determinants of job satisfaction, and the paucity of research on the topic in the Middle East, this study is of great importance to both scholars and practitioners in the region.

The aim of this study is to identify the factors affecting job satisfaction in the UAE. Specifically, the study examines the relative effects of demographic and environmental factors on job satisfaction among Dubai police employees. The study intends to make two important contributions. First, the major limitation of much of the available research on job satisfaction is the predominant focus on individualist Western cultures. This study extends this line of research to the Middle East region focusing on determinants of job satisfaction in the UAE. Second, to overcome the limitations of existing tools for measuring determinants and outcomes of job satisfaction, which have a strong bias towards Western culture[1], we developed and validated a job satisfaction questionnaire designed specifically for UAE organisations. The questionnaire incorporates factors that are reported to contribute to job satisfaction in collectivist countries, such as perceived social status of the job.

The remainder of the paper is structured as follows. The next section provides a brief literature review of the determinants of job satisfaction. In this, the key individual and environmental determinants of job satisfaction are identified and discussed. Thereafter, the method section describes the sampling method and the steps followed to develop the instrument for the study. The analysis section, which follows, presents the findings of the study. The final section of the paper discusses the results of the study and concludes the paper.

Literature review

There is no one definition that sums up job satisfaction. Crudely defined, job satisfaction refers to “the degree to which people like their jobs” (Spector, 1997, p. 7). Scholars use the concept to show a combination of employee feelings towards the different facets of job satisfaction such as the nature of the work itself, level of pay, promotion opportunities, and satisfaction with co-workers (Schermerhorn *et al.*, 2005, p. 158). The importance of studying job satisfaction stems from two important sets of findings. First, job satisfaction is associated with increased productivity and organisational commitment, lower absenteeism and turnover, and ultimately, with increased organisational effectiveness (Ellickson and Logsdon, 2001). According to Wright and Davis (2003), the benefits that employees receive from their organisations influence the effort, skill, creativity and productivity that they are willing to give in return. Organisational interest in job satisfaction has been also motivated by humanitarian interests, namely the notion that employees deserve to be treated with respect and have their psychological and physical well-being maximised (Spector, 1997; Ellickson and Logsdon, 2001). The second important finding is that low job satisfaction has negative outcomes, such as withdrawal behaviour, increasing costs, decreasing profits and, eventually, customer dissatisfaction (Zeffane *et al.*, 2008).

Determinants of job satisfaction

For several decades, numerous studies have attempted to classify and determine factors influencing job satisfaction. The literature on determinants of job satisfaction can be divided into two camps: the content perspective which approaches job

satisfaction from the perspective of needs fulfilment, and the process perspective which emphasises the cognitive process leading to job satisfaction (Foster, 2000; Spector, 1997, p. 6). The content perspective assumes that all individuals possess the same set of needs and therefore prescribes the characteristics that ought to be present in jobs. Content theories include Maslow's (1954) need hierarchy theory and the motivator-hygiene theory proposed by Herzberg *et al.* (1959). Scholars of the process theories, on the other hand, de-emphasise the role of needs, and focus on the cognitive processes leading to job (dis)satisfaction. Process theories include Vroom's (1964) expectancy theory, and Adams' (1963) equity theory. Recent literature (Rollinson, 2008), however, posits that the two perspectives are complementary and advocates that scholars incorporate both of these in the study of those factors that determine job satisfaction.

Research from both needs and content perspectives has reported a plethora of factors in relation to job satisfaction, which can be grouped into two broad categories: demographic factors that focus on individual attributes and characteristics such as sex, age and job level; and environmental factors which pertain to factors associated with the work itself or work environment such as salary, promotion and supervision (Zeffane, 1994; Reiner and Zhao, 1999; Ellickson and Logsdon, 2001). Literature focusing on environmental factors is underpinned by the assumption that job satisfaction is positively correlated with the extent to which individuals' work fulfils their needs (Ellickson and Logsdon, 2001).

Demographic characteristics include factors that define individuals even before their entry into the work situation, such as sex, age and education level as well as other factors related to their work experience, such as job level, shift work, and years of experience. The core assumption here is that given that the workforce of any organisation is not demographically homogeneous, one expects employee satisfaction to differ across sub-groups (Fosam *et al.*, 1998). The small body of research on job satisfaction in Middle Eastern countries suggests a strong association between demographic factors and job satisfaction (Crossman and Abou-Zaki, 2003; Suliman, 2006; Abu Elanain, 2009). In particular, Crossman and Abou-Zaki (2003) reported that female Lebanese employees have higher job satisfaction from fair pay than do male employees. Male employees, on the other hand, obtain more satisfaction from quality of supervision than do female employees.

Besides, demographic factors, much attention has been given to factors arising from the work environment in motivating employees to work with high levels of job satisfaction. A number of scholars, such as Herzberg (1968) and Spector (2008), posit that the work environment has a significant effect on the level of (dis)satisfaction of employees. Over the years, scholars have identified a number of important environmental factors that are thought to influence job satisfaction (Lambert *et al.*, 2001). These include salary level, climate management, promotion opportunities, fairness of appraisal systems, and satisfaction with co-workers.

Although recent researchers advocate that the work environment is a better predictor of job satisfaction (Reiner and Zhao, 1999; Carlan, 2007; Ellickson and Logsdon, 2001; Forsyth and Copes, 1994), a small, but significant, body of research posits that demographic factors are also good predictors of job satisfaction. For instance, researchers have paid significant attention to the association between the sex variable and job satisfaction, albeit reporting inconsistent results (Smith *et al.*, 1969;

Tait *et al.*, 1989). Thus, we consider both environmental and demographic variables in this study.

Research on determinants of job satisfaction across countries provides evidence to suggest that job satisfaction is strongly influenced by national cultural values and norms (Kirkman and Shapiro, 2001; Huang and Van de Vliert, 2004; Roe *et al.*, 2000; Oishi *et al.*, 1999). Generally, employees in Western developed countries have higher satisfaction from intrinsic job factors related to the nature of the work itself than those in developing countries. Research has also shown that employees in non-Western countries derive more job satisfaction from extrinsic job factors such as pay and working conditions than those in Western developed countries (Adigun and Stephenson, 1992; Huang and Van de Vliert, 2004). For instance, research in Western countries shows that mentally challenging jobs lead to job satisfaction in Western countries (Huang and Van de Vliert, 2004; Diener *et al.*, 1995) but not in non-Western countries (Huang and Van de Vliert, 2004; Hui and Yee, 1999). Huang and Van de Vliert (2004) argue that in Western individualist countries employees place a strong emphasis on the job and therefore obtain more satisfaction from carrying out mentally challenging work. In contrast, in collectivist countries where social recognition is often considered more important to life than the satisfaction of personal needs, job satisfaction may be derived from the social perception and status of the job rather than intrinsic job factors (Huang and Van de Vliert, 2004; Diener *et al.*, 1995; Nevis, 1983). Abu Elanain (2009, p. 472) notes that autonomy is not associated with job satisfaction because “UAE culture is high in uncertainty avoidance ... therefore, for UAE people, increasing autonomy is perceived as lack of direction”.

Context of the study

Our data were collected from employees from the Dubai Police Force (DPF). The UAE comprises of seven emirates:

- (1) Abu Dhabi;
- (2) Dubai;
- (3) Sharjah;
- (4) Ajman;
- (5) Umm Al-Qaiwain;
- (6) Ras Al-Khaimah; and
- (7) Fujairah.

The emirate of Abu Dhabi is the capital and the largest of the seven emirates, occupying around 86.7 per cent of the total land area. Dubai is the second largest emirate covering almost 5 per cent of the total area. The population of the UAE in 2005 was estimated to be 4.1 million. The population comprises 20 per cent Emiratis whilst a large proportion (80 per cent) of the people living in the UAE are workers from different parts of the world and include other Arabs, South-Eastern and Southern Asians, and Europeans and Americans (Suliman, 2006). The high percentage of expatriates in the UAE is due to the country's open economic policies and fast-growing economy and the development rate following the discovery of oil (Suliman, 2006).

The UAE is an Arab and Muslim country. The official language is Arabic, but English is well understood and widely used for communication across the country. The social life in the UAE is substantially affected by the values and culture of Islam. All UAE nationals see their religion as an integral part of daily life. UAE society derives its values and ideas mainly from the teachings of Islam and from local culture, traditions and customs. Suliman (2006, p. 64) reported that work values in the UAE are influenced by Islamic values, Arabic tradition, contemporary development policies, as well as generational, social class and gender factors. Similarly, Simadi's (2006) study of young adult Emiratis' values found that religious and cognitive values were the strongest determinants of work values in the UAE.

The DPf was established in 1956 by the Dubai Local Government with a small number of employees. Since then it has displayed steady growth and is now a large organisation with more than 15,000 staff working in 18 departments (Dubai Police, 2009; Abdulla *et al.*, 2008). Given the nature of police work, job satisfaction is a serious issue for this particular public sector organisation. A police force is a service-intensive organisation with a significant proportion of its employees working in direct contact with its customers, the general public. The police, like any other public sector organisation, need to develop and maintain a strong relationship with its human resources in order to effectively perform crime fighting and service provider roles (Abdulla *et al.*, 2008). Researchers have indicated that job satisfaction among police officers is a complex concept and inadequately defined in policing literature (Greene, 1989; Dantzker, 1994; Brunetto and Farr-Wharton, 2003; Pelfrey, 2004; Chan and Doran, 2009). Yet there have been few research studies in this field, and consequently, research effort on job satisfaction in police work is crucial. In calling for greater attention to the organisational environment in police organisations and its influences upon members' job satisfaction, Zhao *et al.* (1999, p. 168) write:

[w]e invite other scholars and practitioners of policing to join in the study of the relationship between work environment and job satisfaction in law enforcement agencies ... to increase the understanding of organisational environment in police agencies and its effects on officers' job satisfaction.

Methodology

The scale development process

The scale development process was conducted in six stages (see Table I for a description of the stages). The first stage began with a comprehensive review of the literature on the determinants of job satisfaction in order to generate a pool of items that reflect the job satisfaction construct. A list of items was gathered to cover all aspects of job satisfaction. In-depth interviews and a focus group were also used to generate additional items. Both techniques assisted in achieving the following goals:

- adding items;
- checking the item clarity;
- investigating the clarity of the item directions;
- deleting repeated items; and
- other suggestions.

State of Scale development		Source of data	Provides evidence for
Stage 1	Constructing a definition and content dimension	Literature review	Understanding the concept of job satisfaction
Stage 2	Generating an item pool	Literature review, in-depth interviews ($n = 40$) and focus group ($N = 6$)	Face validity and content validity
Stage 3	Determining the format for measurement	Literature review, in-depth interviews ($n = 40$) and focus group ($n = 6$)	Reliability (internal)
Stage 4	Judging of items by experts	Experts judges ($n = 3$)	Face validity and content validity
Stage 5	Designing a scale and pilot testing	Pilot study ($n = 9$)	Face validity and content validity
Stage 6	Evaluating and finalising the scale	Survey-DPF ($n = 1,017$)	Dimensionality (factor analysis), reliability (Cronbach alpha)

Table I.
Stages and guidelines in
scale development

Table II shows the numbers of dimensions and items that were used in the scale. Existing examples of survey instruments designed to measure overall job satisfaction include the Job in General Scale (JIG) developed by Ironson *et al.* (1989), and Job Satisfaction Blank Number Five, developed by Hoppock (1935). Examples of survey instruments designed to measure facets of job satisfaction include the Job Descriptive Index (JDI) developed by Smith *et al.* (1969), and the Job Satisfaction Survey (JSS) developed by Spector (1985). The dimensions of the Dubai Job Satisfaction Survey (DJSS) used in this study include nine JSS dimensions and almost all of the five JDI. The DJSS included all dimensions of the MSQ except for three:

- (1) activity;
- (2) achievement; and
- (3) social services.

Component items of achievement, which are related to the task identity dimension, were not retained in the DJSS. The pilot study indicated that task identity, and being active most of the time, were not as important factors affecting employees' job satisfaction in the UAE. Therefore, these items were dropped.

Importantly, and in line with past research on job satisfaction in collectivist cultures, employees involved in the pilot study indicated their satisfaction to be more influenced by the perception of the public towards the services provided rather than by being involved in social services. Therefore, the DJSS included a dimension named "public perception". In addition to the public perception dimension, DJSS introduced new two dimensions that were not included in any of the JDI, MSQ and JSS, namely "performance appraisal", and "job stress".

The generation of the pool of items was followed by determining the format for measurement (stage three). Hinkin (1995) notes that the importance of scaling of items is to achieve sufficient variance or variability among respondents for subsequent statistical analysis. Rating questions most frequently use a Likert-style rating scale, which is believed to be more reliable than the Thurstone scale (Sarantakos, 1988). We used a seven-point Likert scale (1 = strongly agree/satisfied-7 strongly disagree/dissatisfied) to measure the degree of agreement with a specific statement.

To ensure face and content validity, the initial pool of items was given to three academic experts in the field of job satisfaction and Islamic and Arabic culture, together with the construct's definition (DeVellis, 2003) (stage four). The judges were then asked to rate each of the items as either "not at all representative", "somewhat representative", or "clearly representative" of the definition provided. The review, in

Table II.
Dimensions and items in
the scale development
process

Sources	No. of added dimensions	No. of added items	No. of deletion items
Literature review	25	78	–
In-depth interviews	–	36	–
Focus ggroup	–	15	51
Expert panel	–	6	15
Pilot testing	–	2	3
Factor analysis	11	–	3
The final scale	11	65	

addition, included checking the clarity of the dimensions and items, the questions' ranking scale, and the overall layout of the scale. The scale construct was modified and 15 items, which were judged as "not at all representative" or "somewhat representative" were deleted. The remaining 68 environmental factors and nine demographic items, which were considered to have sufficient face and content validity were used for further analysis (see Appendix for list of items).

The face and content validity stage was followed by designing and pilot testing the scale (stage five). This step represents three important aspects:

- (1) designing a scale;
- (2) translating the questionnaire into Arabic; and
- (3) pilot testing.

In addition to the scale for demographic and work environment factors, a sub-scale measuring the general job satisfaction with DPF employees was included in this study. This consists of three items measured on a seven-point Likert style rating scale. These items include: I like my job, My job fulfils my necessary needs, and Overall, I am satisfied with my job. A total score of the General Job Satisfaction variable (GJS), was derived by summing the points for each of the three items. The GJS variable was used to identify rates of job satisfaction among the DPF employees. The GJS also served as a dependent variable in our study.

Since the study was conducted in an Arabic country, the UAE, the questionnaire was translated from English into Arabic. In translating the original English questionnaire into Arabic, particular attention was given to ensure contextual equivalence between two linguistic versions. Therefore, the Arabic version was checked by two native Arabic PhD students, who were fluent in both languages. Finally, before collecting data, all sections of the questionnaire were pilot tested by ten people (three academics, three research students and four DPF employees) to assess the validity of the questions and minimise problems in answering them (Saunders *et al.*, 2007).

After the preceding five stages, the scale was distributed in mid-July 2007 among all DPF employees through the DPF's intranet. The intranet links the departments throughout the DPF. Respondents who did not have access to the Intranet were given hard copies in order to maximise the response rate. A total of 1,027 employees from 18 departments of the DPF completed and returned the questionnaires, from which 1,017 were usable. Therefore, the total response rate was 81.5 per cent.

Data analysis

We ran factor analysis with the 68 environmental factors items following Field's guidelines (Field, 2005). To explore the factor structure, principle components analysis with varimax rotation was used, extracting 11 factors (see Appendix):

- *Factor 1*: salary and incentives.
- *Factor 2*: supervision.
- *Factor 3*: public perception.
- *Factor 4*: promotion opportunity.
- *Factor 5*: organisational policy and strategy.

- *Factor 6:* relationship with co-workers.
- *Factor 7:* professional development.
- *Factor 8:* nature of the work.
- *Factor 9:* communication.
- *Factor 10:* job stress.
- *Factor 11:* performance appraisal.

Each sub-scale consisted of three items or more (range 3-10). According to Isaac and Michael (1990), to have a meaningful factor, at least three items should load on it. Field (2005) suggests deletion of items with factor loading less than 0.4. We removed three items since they emerged as being unrelated to the principle. Further, the reliability of each sub-scale and for the entire scale using the Cronbach alpha reliability test showed high internal consistency with coefficients averaged for the 11 sub-scales ranging from 0.61 to 0.92 and 0.89 for the entire scale (see Table III). The findings indicate that each sub-scale score has adequate internal consistency with sub-scales above the conventional standard of ≥ 0.60 (Field, 2005). The results were listed in factor analysis with Factor 1 accounting for the most variance at 35.09 per cent. Together, the 11 factors accounted for 64.63 per cent of the variability of the 68 items (see Appendix).

Descriptive statistics

Descriptive statistics are presented in Table IV. Respondents were predominantly males representing over 85 per cent of the sample. More than half (58.5 per cent) of respondents were UAE nationals. The majority of the respondents (79.3 per cent) were married. The means of the respondents' age and years of experience were 34.26 and 12.5 years respectively. The majority of the study sample (82.6 per cent) held secondary school certificate and university degrees and only 6.3 per cent of respondents held postgraduate degrees. Nearly two-thirds (66 per cent) were not occupying managerial positions, while 34 per cent of respondents were supervisors or managers. More than half (56.9 per cent) of respondents indicated that they worked in shifts. The majority of

Factors	No of items	<i>n</i>	<i>r</i> *	Sig	Cronbach alpha (α)
Factor 1: salary and incentives	10	985	0.535	0.000	0.919
Factor 2: supervision	7	1,003	0.499	0.000	0.916
Factor 3: public perception	7	1,004	0.466	0.000	0.911
Factor 4: promotion opportunity	6	997	0.527	0.000	0.898
Factor 5: organisational policy and strategy	8	987	0.578	0.000	0.899
Factor 6: relationship with co-workers	6	978	0.534	0.000	0.829
Factor 7: professional development	5	998	0.492	0.000	0.891
Factor 8: nature of the work	7	984	0.484	0.000	0.792
Factor 9: communication	3	1,009	0.380	0.000	0.743
Factor 10: job stress	3	995	0.224	0.000	0.612
Factor 11: performance appraisal	3	1,004	0.429	0.000	0.749
Entire scale	65				0.887

Table III.
The relationship between
environmental variables
and GJS

Note: *Correlation is significant at the 0.01 level (two-tailed)

	Characteristics	<i>n</i>	(%)	Mean	(SD)	Test characteristic (<i>t</i> , <i>F</i> or <i>r</i>) *	df	Sig.
1. Sex	Male	868	85.5	16.75	3.8	1.578	1,013	0.116
	Female	147	14.5	16.14	4.4			
2. Nationality	UAE	594	58.5	16.4	4.1	-2.409	963	0.016
	Non-UAE	422	41.5	17	3.6			
3. Age	968	34.26	8.7	0.164		0.000		
4. Marital status	Single	210	20.6	15.8	4.1	-3.696	1,013	0.000
	Married	807	79.4	16.9	3.8			
5. Educational level	Below secondary school	113	11.1	16.97	4.4	1.082	3	0.356
	Secondary school	485	47.7	16.7	3.7			
	University degree	355	34.9	16.4	4.1			
	Postgraduate degree	64	6.3	16.98	3.6			
6. Job level	Supervisory/managerial	345	33.9	17.2	3.7	3.528	727	0.000
	Non-supervisory	672	66.1	16.4	3.9			
7. Years of experience	990	12.51	8.2	0.185		0.000		
8. Shift work	No	436	43.1	16.8	3.9	0.909	1,007	0.364
	Yes	575	56.9	16.6	3.9			
9. Public contact	No	128	12.6	16.3	4.2	-1.039	1,012	0.299
	Yes	888	87.4	16.7	3.8			

Note: Independent *t*-test was used for difference testing between the two sub-groups. One-way ANOVA (*F*) test was used for the difference between three or more sub-groups. Pearson Correlation test (*r*) was used to measure the correlation between two continuous variables

Table IV.
The association between
demographic variables
and GJS

the study sample (87.4 per cent) indicated that they dealt with the public, and only 12.6 per cent of respondents were not in direct contact with the public.

Relationship between environmental and demographic variables and general job satisfaction (GJS)

The relative influence of the environmental and demographic variables on job satisfaction was examined through bivariate analyses. The results show that five out of nine of the demographic variables were found to have statistically significant relationship with GJS (see Table IV). Higher levels of GJS were associated with each of the following factors:

- holding a non-UAE nationality ($t = -2.409, p < 0.05$);
- increased age ($r = 0.164, p < 0.01$);
- being married ($t = -3.636, p < 0.001$);
- occupying supervisory job levels ($t = 3.528, p < 0.01$); and
- increased years of experience ($r = 0.185, p < 0.01$).

Furthermore, surprisingly, the results reveal that all the 11 environmental factors are significantly correlated with GJS (see Table III).

Regression analyses

Multiple linear regression analyses were conducted to identify the set of the environmental variables that serve as predictors of GJS scores with and without controlling for demographic variables. Table V presents the results from two separate multiple linear regression analyses (Reg 1 and Reg 2) on GJS scores. In the first regression, the eleven environmental variables were entered into the model. In the

Model	Standardised coefficients		
	β (Reg 1) ^a	β (Reg 2)	β (differences)
(F1) Salary and incentives	0.328 ***	0.350 ***	0.022
(F8) Nature of the work	0.268 ***	0.271 ***	0.003
(F3) Public perception	0.267 ***	0.252 ***	0.015
(F5) Organisational policy and strategy	0.243 ***	0.237 ***	0.006
(F2) Supervision	0.208 ***	0.212 ***	0.004
(F6) Relationship with co-workers	0.221 ***	0.208 ***	0.013
(F4) Promotion opportunity	0.179 ***	0.175 ***	0.004
(F7) Professional development	0.102 ***	0.097 ***	0.005
(F11) Performance appraisal	0.120 ***	0.096 ***	0.024
(F9) Communication	0.079 **	0.083 **	0.004
(F10) Job stress	0.090 ***	0.069 **	0.021
Nationality	—	0.102 **	
Sex	—	0.072 *	
Shift work	—	-0.058 *	
Public contact	—	0.052 *	
R ² (%)	46	47	1

Table V.
Results of regression
analyses for determinants
of GJS

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ^aOnly the environmental variables were entered in the model. Both the environmental and demographic variables were entered in the model

second regression analysis we entered both the eleven environmental variables and the nine demographic variables in the model.

Table V presents the standardised coefficient beta (β) for the predictor variables (in order of strength of contribution). Multiple regression analysis on GJS scores while entering both environmental and demographic variables as independent variables yielded a significant model at the 0.002 level with 15 predictor variables accounting for 47 per cent of the variance in GJS. The results showed that salary and incentives have the highest impact on job satisfaction ($\beta = 0.350, p < 0.001$). This was followed by:

- a positive perception of nature of the work ($\beta = 0.271, p < 0.001$);
- public perception ($\beta = 0.252, p < 0.001$);
- organisational policy and strategy ($\beta = 0.237, p < 0.001$);
- supervision ($\beta = 0.212, p < 0.001$);
- relationship with co-workers ($\beta = 0.208, p < 0.001$); and
- promotion opportunity ($\beta = 0.175, p < 0.001$).

Although significant, the following variables had the weakest impact on job satisfaction:

- holding non-UAE nationality ($\beta = 0.102, p < 0.01$);
- a positive perception of professional development ($\beta = 0.097, p < 0.001$);
- performance appraisal ($\beta = 0.096, p < 0.001$);
- communication ($\beta = 0.083, p < 0.01$);
- job stress ($\beta = 0.069, p < 0.01$);
- sex ($\beta = 0.072, p < 0.05$);
- shift work ($\beta = -0.058, p < 0.05$); and
- dealing with public ($\beta = 0.052, p < 0.05$).

A comparison of R^2 and Adjusted R^2 (see Table V) between models one and two, reveals that the demographic variables, namely nationality, sex, shift work, and public contact added only 1 per cent to the total amount of variance explained. In addition, the actual differences in the regression coefficient (β) values for the 11 environmental variables between the two models were very small and appear to be of little significance. From the above findings, one can conclude that the set of the eleven environmental variables is a stronger predictor of GJS than demographic variables.

Discussion

The first objective of this paper is to identify the determinants of job satisfaction among the DPF employees in the UAE. We employed regression analysis to capture the relative influence of demographic and environmental variables on job satisfaction. Taken together, our findings indicate that environmental factors are the key determinants of job satisfaction. In line with past research that indicates that demographic variables are of little value to understanding job satisfaction (e.g. Carlan, 2007; Ellickson and Logsdon, 2001; Zhao *et al.*, 1999; Forsyth and Copes, 1994), our results show that demographic variables added only 1 per cent to the total amount of variance explained in the regression model. This finding is important even though it is

not original. It helps future research focused on environmental factors to understand variations in the level of job satisfaction in UAE organisations. Based on this finding, academics and practitioners should concentrate on the nature of the job rather than on the characteristics of the individual to understand why some employees express high level of satisfactions with their jobs and others report lower levels of satisfaction in UAE organisations.

Another equally important objective of the study is to develop a scale of job satisfaction for use in UAE organisations. Accordingly, the scale development process undertaken in this study has resulted in the construction of a multi-dimensional scale, which has demonstrated its validity, and reliability. Unlike previous scales, which were developed in Western cultures, this research attempted to construct a more suitable scale measuring the major dimensions of job satisfaction of employees in a collectivist culture. The developed scale incorporates factors that are specific to collectivist cultures, such as external perception of the job.

Of the 11 work environment factors, salary and incentive clearly emerged as the most powerful determinant of job satisfaction. The finding suggests that DPF workers pay a great deal of attention to remunerations such as salary, fringe benefits, allowances, recognition, and financial rewards. Remunerations can mean different things in different contexts. While salary can be seen by a significant proportion of employees in rich Western countries as a source of recognition (Locke, 1976), for workers in emerging economies or lower paid workers in developed countries, salary can mean security (Al-Saadi, 1996). We reason that employees put a strong emphasis on salary and incentives in the UAE because of their impact on living standards and their importance in providing a sense of security (Aksu and Aktas, 2005), especially for non-UAE nationals where perceived job security is very low (Budhwar and Mellahi, 2007). Furthermore, the high importance paid to salaries and incentives could be due in part to the high cost of living in Dubai compared to other Middle Eastern emirates (Bagaeen, 2007), where for instance residential rents rose sharply in the last few years reaching up to 50 per cent in 2006 (Westley, 2006).

In addition to salary and incentives, the findings show that satisfaction with the job requires satisfaction with different aspects of the work, including variety in the skills and activities required to carry out the tasks, and autonomy in performing work. This is because work itself might provide opportunities for creativity and task variety, allowing employees to increase their knowledge and change in responsibility (Hackman and Oldham, 1975). Although, our results are consistent with past studies in Western countries that showed one the most important influences on a person's job satisfaction experience to emanate from the nature of the work (e.g. Herzberg *et al.*, 1959; Hackman and Oldham, 1975; Judge and Church, 2000; Wong *et al.*, 1998; Luthans, 2005), they challenge the widely held assumption that in collectivist cultures employees have low satisfaction from intrinsic job factors related to the nature of the work itself (Adigun and Stephenson, 1992; Huang and Van de Vliert, 2004; Diener *et al.*, 1995).

Public perception was the third strongest predictor of job satisfaction. This finding shows that respondents thought that they were viewed favourably by the public, and this good image reflects the effective services provided. This result lends credence to past research on job satisfaction in collectivist cultures where employees have high

satisfaction from the social perception and status of the job (Huang and Van de Vliert, 2004; Abu Elanain, 2009).

Our results show that organisational policy and strategies is the fourth strongest predictor of job satisfaction. Organisational policy and strategies includes setting rules and regulations, having moral values, being less bureaucratic, focusing on internal and external customers, and the provision of job descriptions. According to Locke (1976), clear policies and procedures play an important role in helping employees to understand organisational goals; in contrast ambiguity can be a great source of dissatisfaction and frustration. Abu Elanain (2009) has recently argued that UAE employees prefer clear goals and objectives, well-defined lines of authority, and less autonomy because of their high levels of uncertainty avoidance.

The results of this study reveal that relationship with co-workers is the fifth strongest determinant of job satisfaction. This finding reflects the importance that social relations in the workplace can have on employee job satisfaction. Overall, employees in the DPF who reported having supportive and constructive relationships with their team members, also reported higher levels of job satisfaction. This could result from the emphasis placed by Islam on strong relationships among groups, and the high tendency towards collectivism in UAE society. This result is in line with the extant research in collectivist cultures where employees are reported to put a strong emphasis on co-operative and collegial work (Huang and Van de Vliert, 2004).

Expectedly, our results show supervision to be a significant determinant of job satisfaction. This is understandable in high power distance societies, such as the UAE, where supervisors play a major role in the quality of the working lives of subordinates (Abu Elanain, 2009; Kirkman and Shapiro, 2001; Huang and Van de Vliert, 2004; Diener *et al.*, 2003). Also, and consistent with research in Western countries (Ting, 1997; Fosam *et al.*, 1998; Ellickson and Logsdon, 2001), our results show that promotion opportunity is a significant determinant of job satisfaction in UAE organisations. Vroom (1982) argues that promotional opportunity is a goal most workers desire and that an individual's performance is related to the degree to which the individual believes that being promoted is related to performance on the job and how strongly the individual desires the promotion.

Another factor linked to power distance is performance appraisal since supervisors are generally reluctant to engage in two-way communication in the appraisal process (Chang *et al.*, 2009; Fletcher, 2001). Our study reveals that the perception of equity in appraising employee performance is linked to job satisfaction. Past research on performance appraisal in the Middle East argues that the method by which performance appraisal is carried out determines the outcomes of the appraisal system (Crossman and Abou-Zaki, 2003). This is because in Middle Eastern societies, particularistic factors such as family and social connections may shape the outcome of the appraisal process (Budhwar and Mellahi, 2007).

Interestingly, and perhaps surprisingly, our results show that although personal development is a significant determinant of job satisfaction, it is not one of the strongest predictors. One possible explanation for this result is perhaps the overwhelming level of training provided by the DPF to its employees, and as a result personal development is taken for granted. Expectedly, communication, which is reported to be a key predictor of job satisfaction in general, and in police forces in Western countries (Fosam *et al.*, 1998), is the second weakest predictor of job

satisfaction in our study. Again, this could be due to the high power distance culture in which top-down decisions are generally expected and accepted from those with high power and status.

The most surprising result is the role of job stress in determining job satisfaction. Although, our results confirm its significance in predicting job satisfaction, its relative importance is very low. This is surprising given that working for the police is one of the most stressful and dangerous jobs where employees encounter many stressful experiences related to the nature of the profession, involving critical moments such as risk-taking and engagement with criminals (Davey *et al.*, 2001).

Overall, our findings show that in a collectivist culture such as the UAE, both intrinsic and extrinsic factors can be a source of job satisfaction or dissatisfaction. In contrast to past research which argues that employees in collectivist countries derive more job satisfaction from extrinsic factors such as social perception of the job than intrinsic factors such as task variety (Adigun and Stephenson, 1992; Huang and Van de Vliert, 2004; Diener *et al.*, 1995; Hui and Yee, 1999), our results suggest that employees in the UAE consider both sets of factors as important. In line with the extant research on job satisfaction in non-Western countries, our results show that salary and incentives are the strongest determinants of job satisfaction.

Our study's results should be viewed in light of our research design's limitations, which also serve to highlight future research opportunities. Our focus on one public sector organisation based in the Emirate of Dubai limits the generalisability of our results. Although our results seem likely to apply directly to other public organisations in the UAE, studies involving private and public sector organisations are required in order to establish that environmental factors are the key predictor of job satisfaction in UAE organisations. This is necessary because a number of scholars (Vasu *et al.*, 1998) reported that the determinants of job satisfaction tend to be different in private and sector organisations.

A second limitation is that our data is collected from a single Emirate, i.e. Dubai. And even though Dubai shares similar cultural factors and economic backgrounds with other emirates in the UAE and neighbouring Middle Eastern countries such as Qatar, Kingdom of Saudi Arabia, Bahrain, Kuwait and Sultanate of Oman, one nonetheless, expects significant differences between them. This highlights opportunities for future inquiry to examine the determinants of job satisfaction in other emirates in the UAE and other Middle Eastern countries.

Conclusion

Understanding the determinants and consequences of job satisfaction has been a key task of industrial-organisational psychology and organisational behaviour scholars for several decades, but received little attention in the Middle East. This study takes an important step forward by identifying the key determinants of job satisfaction in the UAE. The study examines the association between job satisfaction and environmental and demographic factors. Our results reveal that environmental factors are better predictors of job satisfaction than are demographic factors. In addition, our results reveal that salary and incentives are the most important determinants of job satisfaction. In contrast to the extant research which posits that employees in collectivist cultures derive more job satisfaction from factors extrinsic to the job, our findings suggest that both intrinsic and extrinsic factors can be a source of job

satisfaction. Our hope is that these findings are tested in other public and private organisations in the UAE and wider Middle East to further uncover how different factors result in job satisfaction, and thereby help develop a more complete understanding of the determinants of job satisfaction in the Middle East.

Note

1. We discuss the 11 environmental variables in the order of their relative strength of association with job satisfaction.

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Appendix

Determinants of job satisfaction

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Items	Factors
<i>Factor 1: salary and incentives</i>	
My salary is adequate for my living expenses	0.831
The period between pay rises is reasonable	0.824
I feel appreciated by the organisation when I think about what they pay me	0.810
My organisation has an appropriate salary scale	0.787
All necessary fringe benefits are provided in my organisation (e.g. health insurance, travel tickets, accommodation and allowances)	0.694
The performance recognition system in my organisation is as good as in most other organisations	0.632
There are adequate financial rewards for those who work here	0.546
Regarding salary, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.498
Regarding fringe benefits, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.482
Regarding recognition and rewards, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.412
<i>Factor 2: supervision</i>	
My supervisor is available when needed	0.798
My supervisor shows consideration for subordinates' feelings	0.794
My supervisor gives me the opportunity to participate in important decision-making	0.776
Communication between me and my immediate supervisor is good	0.754
I receive regular feedback about my performance	0.720
The feedback I receive from my supervisor is useful	0.635
The form my supervisor uses accurately evaluates my performance	0.436
<i>Factor 3: public perception</i>	
Crime control	0.830
The presence of police officers in public places	0.801
Traffic safety	0.791
Community educational programs	0.790
Response time	0.746
Communications skills (e.g. attitude, concern and follow-up)	0.722
Professionalism (e.g. problem solving)	0.650
<i>Factor 4: promotion opportunity</i>	
My organisation has a clear career path (that is, the defined track a person follows in the pursuit of professional goals)	0.800
Promotion opportunities are not limited and are fairly adequate in this organisation	0.778
My organisation has a clear and fair promotion policy and strategy that takes efficiency, performance and experience into account	0.676
My organisation puts the right person in the right position	0.543
My organisation gives proper attention to staff complaints and grievances	0.521
Regarding promotion, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.413
<i>Factor 5: organisational policy and strategy</i>	
The procedures in my organisation do not interfere with good job performance	0.720
The general policy for my organisation is congruent with moral values	0.709
	(continued)

Table AI.
Factorial analysis for job
satisfaction factors

Items	Factors
This department is doing a good job of providing steady employment	0.673
The organisation takes proper precautions to ensure a safe and healthy workplace	0.608
The working conditions (e.g. air conditioning, lighting and workplace) are comfortable	0.541
My organisation gives considerable attention to public needs (e.g. running annual surveys)	0.496
My organisation gives proper attention to complaints by the public	0.464
My organisation has a clear strategy to improve public service quality delivery	0.448
<i>Factor 6: relationship with co-workers</i>	
I receive an assignment with the proper staffing to complete it	0.662
I work with a team who operate quite consistently	0.597
I do not receive incompatible requests from all of my supervisors	0.542
Honest competition exists between workers in this organisation	0.505
There is no bickering and fighting at work	0.486
I do not have difficulty in my job because of the competence of the people with whom I work	0.415
<i>Factor 7: professional development</i>	
The organisation provides me with enough work-related training	0.799
The organisation provides me with adequate resources and chances to develop myself professionally (e.g. workshops, courses and conferences)	0.777
My organisation practises job rotation (that is, the rotation of workers between different tasks to acquire skills)	0.600
I am given the chance to try out some of my own ideas	0.502
Regarding opportunities for professional development, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.492
<i>Factor 8: nature of the work</i>	
My work is varied and is not routine	0.725
My job requires the use of a number of skills	0.707
The work I do contributes to the organisation's goals	0.682
A lot of other people can be affected in some way by my work	0.637
My job gives me the opportunity to work independently	0.582
My job requires me to be very creative	0.547
My job goals are clear	0.522
<i>Factor 9: communication</i>	
The equipment required for effective communication is widely available in this organisation (e.g. telephone, fax and internet)	0.670
Information is shared openly within this organisation (e.g. bulletins, circulars and orders)	0.611
Communication between me and other workers is good	0.562
<i>Factor 10: job stress</i>	
My job does not require me to work extra hours	0.796
Work assignments are not out of my specialised field	0.657
My job does not interfere with my family and social life	0.581
<i>Factor 11: performance appraisal</i>	
My performance appraisal is conducted on time each year	0.587
I always receive a copy of my annual performance appraisal	0.431
Regarding performance appraisal, I feel that I am treated fairly compared with colleagues in my organisation who have similar qualifications and who have served a similar number of years	0.414

Table AI.

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